

Ajay Rajendra Kumar

Boston, MA, USA

ajayrajendrakumar8@gmail.com

Website

GitHub

LinkedIn

Google Scholar

Education	Northeastern University , Boston, USA	Sep. 2024 – May 2026
	M.S. in Computer Science - GPA: 4.0/4.0 - Capstone: <i>Dense Visual SLAM on Edge Devices</i> - Advisor: Prof. Huaizu Jiang	
	Manipal Institute of Technology , Manipal, India	Aug. 2020 – Jul. 2024
	B.Tech. in Computer Science and Engineering - GPA: 9.3/10.0 - Minor: Computational Intelligence - Dissertation: <i>Evaluating Deep Learning Approaches for Fracture Segmentation in X-ray Images</i> - Advisor: Prof. Subhamoy Mandal (IIT, Kharagpur, India)	
Research Experience	FarSight Lab, Northeastern University	Nov. 2025 – Jun. 2026
	Research Assistant with Prof. Lorenzo Torresani, Boston, USA - Improved video question answering performance of PerceptionLM-8B by 12% on the EgoBlind benchmark through prompt optimization and supervised fine-tuning; identified abstention on safety-related questions limited the model's practical utility. - Developed a bottom-up visual grounding pipeline that tracked visible entities and generated question-relevant textual evidence; found that even accurate evidence rarely changed incorrect model responses, revealing unreliable evidence integration. - Improved PerceptionLM-8B performance on V-STaR by 9% and 14% in zero-shot and LoRA fine-tuned settings, respectively, using question-relevant visual guidance through SAM3-generated bounding-box overlays; investigated training a question-conditioned visual grounding module with reinforcement learning to automate visual guidance. - Designed structured chain-of-thought supervision requiring models to identify relevant visual evidence before reasoning toward an answer; constructed 10K+ grounded training traces for parameter-efficient fine-tuning; found that models could learn to cite visual evidence without reliably grounding their reasoning in the provided regions.	
	Visual Intelligence Lab, Northeastern University	Apr. 2025 – Mar. 2026
	Research Assistant with Prof. Huaizu Jiang, Boston, USA - Identified fundamental limitations of sparse voxel-based visual odometry through systematic ablation studies, tracing pose estimation errors to geometric information lost at low voxel resolutions and motivating a pivot toward dense correspondence-based SLAM. - Designed and trained a lightweight uncertainty estimation module to predict pixel-wise correspondence confidence, filtering unreliable matches to improve the robustness and accuracy of downstream camera pose estimation. - Developed a scalable Kubric-based synthetic data generation pipeline to render 2K+ stereo video sequences with diverse camera trajectories, object configurations, and HDRI environments for training the confidence estimation network. - Co-developed NeuSLAM, a dense visual SLAM system integrating the confidence-guided front-end with an efficient classical pose-tracking back-end, achieving 9.9% F1-all on KITTI 2015 optical flow and competitive zero-shot synthetic-to-real stereo disparity performance on KITTI 2015 and ETH3D.	
	D'Amore-McKim School of Business, Northeastern University	Feb. 2025 – May 2026
	Research Assistant with Prof. Anna Lamin & Prof. Valentina Marano, Boston, USA - Developed an end-to-end data acquisition pipeline to collect and preprocess corporate social responsibility disclosures from Indian companies, curating 100K+ records spanning five years for empirical analysis. - Developed scalable Python-based ETL pipelines for JSON and PDF documents, incorporating structure-aware document chunking, metadata extraction, and automated preprocessing to support large-scale information retrieval and extraction.	

- Developed an LLM-powered document intelligence pipeline combining semantic retrieval, targeted OCR, and OpenAI models to extract, normalize, and structure information from unstructured corporate reports for downstream empirical analysis embedding-based retrieval and GPT-4.1 Likert scoring to evaluate 10K+ Crunchbase company profiles against ten research criteria, enabling analysis of the domestic market orientation of Indian firms.

Manipal Institute of Technology

Jan. 2023 – Jul. 2023

Research Assistant with Prof. Nisha P. Shetty, Manipal, India

- Developed and evaluated misinformation classification models in PyTorch, comparing transformer-based architectures including XLNet and BERT on the COVID-19 ANTi-Vax dataset of 15K+ tweets.
- Collected and processed community discourse from targeted vaccine-related subreddits using the Python Reddit API Wrapper (PRAW) for downstream social media analysis.
- Developed a user-association mapping algorithm based on cosine similarity and constructed interaction networks to analyze relationships among users using NetworkX.

Publications

* denotes equal contribution

INTERNATIONAL CONFERENCE

1. Aniket Gupta, Tianye Ding, **Ajay Rajendra Kumar**, Dennis Giaya, Pia Bideau, Charles Saunders, Qu Cao, Aruni RoyChowdhury, Hanumant Singh, and Huaizu Jiang. *NeuSLAM: Dense Visual SLAM on Edge Devices*. **Spotlight** at ICRA 2026 Workshop on From Sea to Space: Advancing Perception in Harsh Domains; accepted to CVPR 2026 Demo Track.
2. C. Rao*, G. M. Prabhu*, **Ajay Rajendra Kumar***, S. Gupta, and N. P. Shetty. *Addressing Vaccine Misinformation on Social Media by Leveraging Transformers and User Association Dynamics*. International Conference on Machine Learning and Data Engineering (ICMLDE), 2023.

Industry Experience

Fidelity Investments, Bengaluru, India

Jun. 2023 – Aug. 2023

Software Development Intern

- Reduced dashboard query latency by 90% by migrating the underlying data warehouse from Oracle to Snowflake and re-architecting query execution for large-scale change-ticket analytics.
- Delivered full-stack features for an internal change-ticket management platform using Angular and Spring Boot, redesigning core workflows for creating, tracking, and managing engineering change requests.
- Designed and integrated RESTful APIs using Spring Boot to connect the Angular frontend with Snowflake-backed backend services, enabling efficient ticket creation, tracking, and reporting workflows.

Grad Course Projects

Pose Graph Sparsification via Algebraic Connectivity

Mar. 2026

MOBILE ROBOTICS, Northeastern University

- Developed pose graph sparsification algorithms based on Maximizing Algebraic Connectivity (MAC) to select informative edges while reducing graph size for long-horizon SLAM.
- Implemented iterative MAC-based edge selection with continuous relaxation and Madow/nearest-neighbor rounding, using odometric backbones and minimum spanning trees to preserve graph connectivity.
- Evaluated sparsification methods across standard SLAM pose graph datasets, including KITTI, Intel, Sphere, and City datasets.

Equivariance for Text-Driven Human Motion Generation

Mar. 2025

DEEP LEARNING, Northeastern University

- Integrated equivariant neural networks into a diffusion-based Motion Diffusion Model (MDM) to enforce rotational and reflectional symmetries for text-driven human motion generation.
- Modified the MDM Transformer encoder with EquiLinear layers using scalar irreducible representations to preserve motion symmetries during generation.
- Trained and evaluated the model on HumanML3D using CLIP text embeddings, measuring generation quality and text-motion alignment with FID and R-precision.

Teaching Experience

Object-Oriented Design (CS2163)

Aug. 2023 – Dec. 2023

Undergraduate Teaching Assistant, Manipal Institute of Technology

- Assisted 20+ undergraduate students with object-oriented programming, software design principles, and Java programming through weekly office hours and hands-on guidance.
- Supported evaluation of programming assignments and provided feedback on code quality, design, and implementation.

	Problem Solving Using Computers (CS1081)	Aug. 2023 – Dec. 2023
	Undergraduate Teaching Assistant, Manipal Institute of Technology	
	• Guided 20+ undergraduate students in programming fundamentals, algorithmic problem solving, and debugging through weekly lab sessions and individual support. • Evaluated programming assignments and provided feedback to help students improve their C programming skills.	
Leadership	Indian Society for Technical Education (ISTE), Manipal Chapter	Jun. 2022 – Dec. 2023
	Machine Learning Head, Manipal Institute of Technology	
	• Led the chapter's machine learning initiatives, coordinating a team of 30+ students and organizing technical workshops, seminars, and hands-on sessions. • Mentored 3 student teams on machine learning projects, providing guidance on problem formulation, model development, experimentation, and evaluation. • Collaborated with student organizers and faculty members to develop the chapter's technical programs and expand student engagement in machine learning.	
Open Source	TorchMetrics , Contributor	2026
	• Fixed a Transformers compatibility issue in CLIPScore by handling dataclass-based feature outputs and adding regression tests; merged upstream (#3408).	
Technical Skills	Programming: Python, C/C++, Java, SQL, R Frameworks & Tools: PyTorch, TensorFlow, vLLM, HuggingFace, PEFT (LoRA, QLoRA), bitsandbytes, Weights & Biases, Git Distributed Training: PyTorch Distributed (DDP/FSDP), Slurm, DeepSpeed Web & Data: Angular, Spring Boot, Snowflake, Oracle DB	
Honors & Awards	Mar. 2025 Most Innovative Project Award, Deep Learning (CS7150)	Boston, USA
	Oct. 2023 Undergraduate Research Sponsorship Grant	Dehradun, India
	Aug. 2022 National 2nd Runner-Up, Daimler India Hackathon	Chennai, India
	Apr. 2022 1st Runner-Up, Transfusion Medicine Hackathon	Manipal, India
	Jun. 2021 Winner, Space Odyssey Web Development Hackathon	Manipal, India
	Jan. 2021 Achiever's Scholarship Award	Manipal, India
Community Involvement	Child Rights and You (CRY), <i>Outreach Volunteer</i>	Mar. 2023 – Aug. 2024
	AIIESEC, <i>Outgoing Global Talent Department, Mentor</i>	Mar. 2023 – Sep. 2023
	Indian Association of Physiotherapists, <i>COVID-19 Chatbot Developer</i>	May 2021 – Jun. 2021